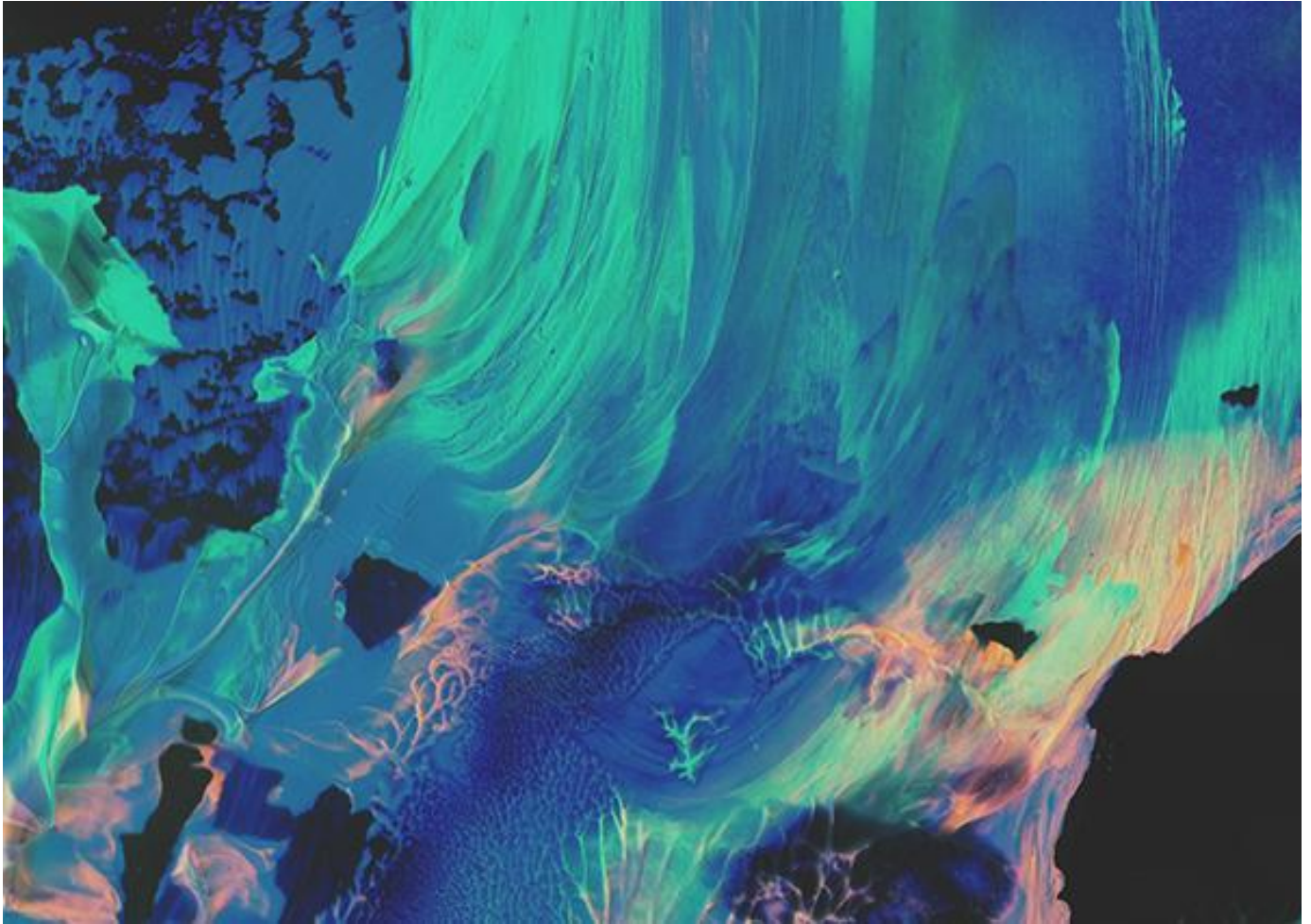




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Survival Models For Dementia Risk Prediction

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Abstract

Advances in healthcare systems have allowed people to live longer than ever before, leading to a rise in age-related diseases such as dementia. Accurate risk prediction models may allow for early identification of individuals at increased risk. This might enable efficient preventive interventions before the onset of irreversible cognitive decline. In this thesis, risk factors for Alzheimer's disease and vascular dementia were investigated using survival analysis. The study included over 400,000 individuals from the UK Biobank after filtration. Cox proportional hazards regression was used to assess the model assumptions, and cross-validated lasso Cox regression was employed to identify the most informative predictors. The analyses showed that the Cox models satisfied the assumptions of proportional hazards and linearity. Lasso regression suggested that Alzheimer's disease could be predicted using only the APOE genotype, age and education, whereas vascular dementia was predicted using APOE, age, diabetes, ischemic heart disease and hypertension. The project was affected by an unexpected UK Biobank research platform shutdown, preventing completion of the planned model evaluation and risk chart development. Nevertheless, the study provides important insights for future work on dementia risk prediction models.

Author contributions

The study was based on data obtained from the UK Biobank after project application approval. Data preprocessing was performed by Nicolai Sandu. The overall project framework was established together with the supervisor. Within this framework the author performed the literature review, implementation of statistical methods, model development, programming, statistical analyses, generation of all figures and tables (unless specified), interpretation of results, and thesis writing. The supervisor provided scientific guidance and feedback throughout the project.

Introduction

Dementia is a syndrome characterized by a decrease in cognitive abilities, problems with memory, thinking, behaviour, and motor control (World Health Organization, 2025). Currently more than 55 million people live with dementia worldwide and it is estimated that 150 million will be affected by 2050 due to ageing populations (Collaborators, 2022). This will result in an ever-increasing burden on public healthcare systems, elderly care and informal caregivers. Informal caregivers alone are estimated to spend on average more than five hours of care and supervision per day (Neubauer et al., 2008). This unpaid work was valued at \$346.6 billion in 2023 in the United States alone (“2024 Alzheimer’s Disease Facts and Figures,” 2024). No cure or effective treatment is available; effective preventive strategies are necessary. (Birks & Harvey, 2018; Cohen et al., 2023; Matsunaga et al., 2015)

Previous studies have identified 16 potentially modifiable risk factors: length of education, hypertension, hearing loss, smoking, obesity, depression, diabetes, low social contact, excessive alcohol intake, traumatic brain injuries and air pollution (Livingston et al., 2024). The 16 risk factors are estimated to account for around 40 % of all dementia cases, meaning that up to 40 % of dementia cases might have been delayed or prevented entirely if behavioural and environmental changes were applied earlier in life (Livingston et al., 2024).

10-year risk estimators in the form of risk charts have been developed for incident all-cause dementia (Juul Rasmussen et al., 2020; Zheng et al., 2023). These risk charts are intended to identify individuals at high risk of dementia. Dementia, however, is caused by several different neurodegenerative diseases: Alzheimer's disease (AD) (60–70% of cases), vascular dementia (VD) (15–30% of cases) and others (World Health Organization, 2025). The etiology of AD is largely unknown; however, the neurodegeneration is associated with the buildup of peptide deposits of beta-amyloid and phosphorylated-tau proteins. AD starts with hippocampal atrophy, followed by atrophy in the prefrontal and temporal cortex; in the latter stages, it spreads to the rest of the brain (Abdulkhaliq et al., 2026). VD is defined as cognitive decline in the presence of brain damage due to impaired blood flow or damage to blood vessels following one or several cerebrovascular events, such as lacunar infarcts, large infarcts, white matter lesions, small vessel disease or intracranial haemorrhages. Depending on the nature of the cerebrovascular event, dementia caused by VD can vary in rate of onset (Korczyn et al., 2012). The etiological difference between AD and VD calls for more risk-specific charts that can provide disease-specific predictive capabilities.

Aim

The aim is to develop 10-year risk charts for AD and VD building on the work of Minois-Genin (Minois-Genin, 2025). The project aims to include an updated version of the UKB, test new covariates, investigate interactions between covariates, perform a rigorous check of the Cox-model assumptions and improve the performance of the Cox-model through selection of covariates using cross-validated lasso regression. This robust dementia-risk-prediction model will be used to create new risk charts, intended for use by medical professionals to provide statistically supported insights and advice to people at high risk of dementia.

Materials and methods

UK Biobank

The data used in this study come from the UK Biobank (UKB) with over 500,000 voluntary participants. Participants were aged 40–69 years at recruitment between 2006 and 2010 from across the United Kingdom (UK). Data were collected through blood sampling, physical examination, an elaborate questionnaire and various organ-function scans. The participants consented to the UKB linking their hospital records with continuous updates. The data used in this study have 16–20 years of follow-up. More than 18,000 peer-reviewed scientific papers are related to the UKB (“Publications,” 2026).

The genetic data were extracted from blood samples on which whole-genome sequencing was performed. Genetic variants were extracted from the imputed genetic data. Ancestry was determined using the genetic data, and only participants of European descent were included in the downstream analysis.

Due to the depth of individual detail in the data and risk of re-identification of participants, the privacy concerns are paramount (Bycroft et al., 2018; Sudlow et al., 2015). Data security measures include transfer and storage encryption, an online data access platform, data anonymization. The data access application processes include courses in cybersecurity, approval of a research plan and proof of being part of an established team of researchers. The data security measures are enforced through termination of access in case of suspicion of security breaches (“Apply for Access,” 2026).

On 23 April 2026 the UKB was shut down due to a leak in China (Nightingale, 2026). This resulted in a zero-notice data access termination for all UKB-based research projects across the world, including this one. As a result, the final steps of analysis as well as corrections to the figures could not be completed even though the analysis pipeline was proceeding on schedule.

Endpoint definitions

AD and VD are defined as diagnoses made during follow-up; the definitions are presented in Table 1. Vascular-related dementia is a constructed category of VD and other unspecified dementia. Unspecified dementia is diagnosed when the practitioner cannot specify what type of dementia a patient has and it can therefore represent any type or several simultaneously. This category can be used to provide greater statistical power for assessing the impact of cardiovascular and cerebrovascular diseases on VD, since mixed dementia is likely to include some cases of VD. Table 1 shows the International Classification of Diseases (ICD) for endpoints of AD, VD and VRD (*ICD-9-CM Lookup - Explore Legacy Diagnosis Codes*, n.d.; *ICD-10-CM*, n.d.).

Table 1: International Classification of Diseases codes (ICD), ICD-9 and ICD-10 codes for endpoint definitions.

Diagnosis	ICD-9	ICD-10
Alzheimer’s disease	3310	Foo, Goo
Vascular dementia	F01	F01

Vascular-related Dementia	2904, 2942	FO1, FO3
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Covariate definitions

All covariate values are recorded at the study start and not modified throughout follow-up time.

Risk Factors - Inherent

Age is defined as the time from birth until entry into the UKB. Dementia is not a normal part of ageing, although it is highly correlated with age. The risk for dementia doubles approximately every five years for ages >65, making age the greatest known risk factor. ("2024 Alzheimer's Disease Facts and Figures," 2024). Aging is a biological and chemical process driven by many different processes (López-Otín et al., 2023), many of which may be affected by one's lifestyle.

Apolipoprotein E (APOE) mediates lipid transport in the bloodstream and in the brain thus affecting various functions throughout the body in different ways. One of the genetic variants of the protein, the $\epsilon 4$ variant alone, is one of the strongest known risk factors for AD (Raulin et al., 2022). The $\epsilon 3$ variant is the wild-type and the $\epsilon 2$ variant may have a protective effect. Generally, the following 6 APOE genotype categories are considered common enough to be studied: $\epsilon 22$, $\epsilon 23$, $\epsilon 33$, $\epsilon 24$, $\epsilon 34$, $\epsilon 44$.

A polygenic risk score is a derived risk score based on a number of genes and their effect on the outcome. The polygenic risk score used in this analysis was created by a weighted sum based on the number of alleles present for an individual. The genes were identified in a genome-wide association study (GWAS) on AD by Bellenguez et al. (Bellenguez et al., 2022). The weight applied on each allele is the negative logarithm of the odds ratio from a logistic regression performed in the GWAS, assuming that the risk is additive, meaning that the risk is assumed to double if an allele is present on both strands.

Previous studies have shown that female sex is associated with a higher risk of AD. However, this trend was not statistically significant for most ages, and the difference is likely explainable by historically lower education among females (Huque et al., 2022). The sex covariate corresponds to genetic sex, where the reference value (o) is female.

Risk Factors - Modifiable

Smoking is defined by UKB Data-Field 20160 where participants are classified according to whether they had ever smoked. Smoking is associated with a range of negative health side effects (Hernán et al., 2008; Yanbaeva et al., 2007). The side effects in combination with the modifiability make it a useful risk factor to include for possible preventive use.

Alcohol intake is classified as high if the questionnaire answer exceeds "one or three times a month" based on UKB Data-Field 1558. Upon ingestion, ethanol is converted into acetaldehyde. Acetaldehyde is the primary intoxicant when forming adducts with dopamine, which may contribute to alcohol use disorder. Acetaldehyde may also form adducts with DNA, which has genotoxic effects (Zakhari, 2006), in turn contributing to an acceleration of biological age which is highly associated with dementia (Zahr, 2024).

Low education is defined as having answered "none of the above" in UKB Data-Field 6138. Cognitive reserve, which can be built up over years of education, has been identified as providing resistance to dementia. It allows neuronal degeneration to occur without giving rise to noticeable symptoms. (Meng & D'Arcy, 2012).

Blood pressure is defined as UKB Data-Field 4079 for diastolic blood pressure and UKB Data-Field 4080 for systolic blood pressure.

Intracranial stroke, diabetes, hypertension and ischemic heart disease are defined in Table 2.

Table 2: ICD-9 and ICD-10 codes for comorbidity covariate definitions. Covariates in the table are defined at baseline and not modified throughout follow-up time.

Diagnosis	ICD-9	ICD-10
Intracranial stroke	433, 434 & 431	I63 & I61
Ischemic Heart disease	410–414	I20–I25
Diabetes	250	E10, E11, E13, E14
Hypertension	401	I10, I11, I12, I13, I15

Analysis flow

Initially, familiarity was established with the covariates which was previously been identified as being related to dementia. This was done through the creation of a table of descriptive statistics with median, standard deviation, sample count, missing values and more for each covariate. Additionally, the assumptions of proportional hazards and log hazard linearity were assessed for all covariates. Next, interaction terms in the Cox models were investigated. After that, Cox lasso regression with a cross-validation was performed to identify which covariates are the most important for prediction. Then, cumulative incidence was estimated with competing risks using Fine-Gray risk regression to create the final risk values. Using different covariate subsets obtained from the cross-validation step, the prediction performance and generalizability of the subsets were compared. Finally, risk charts were created for application in prevention settings.

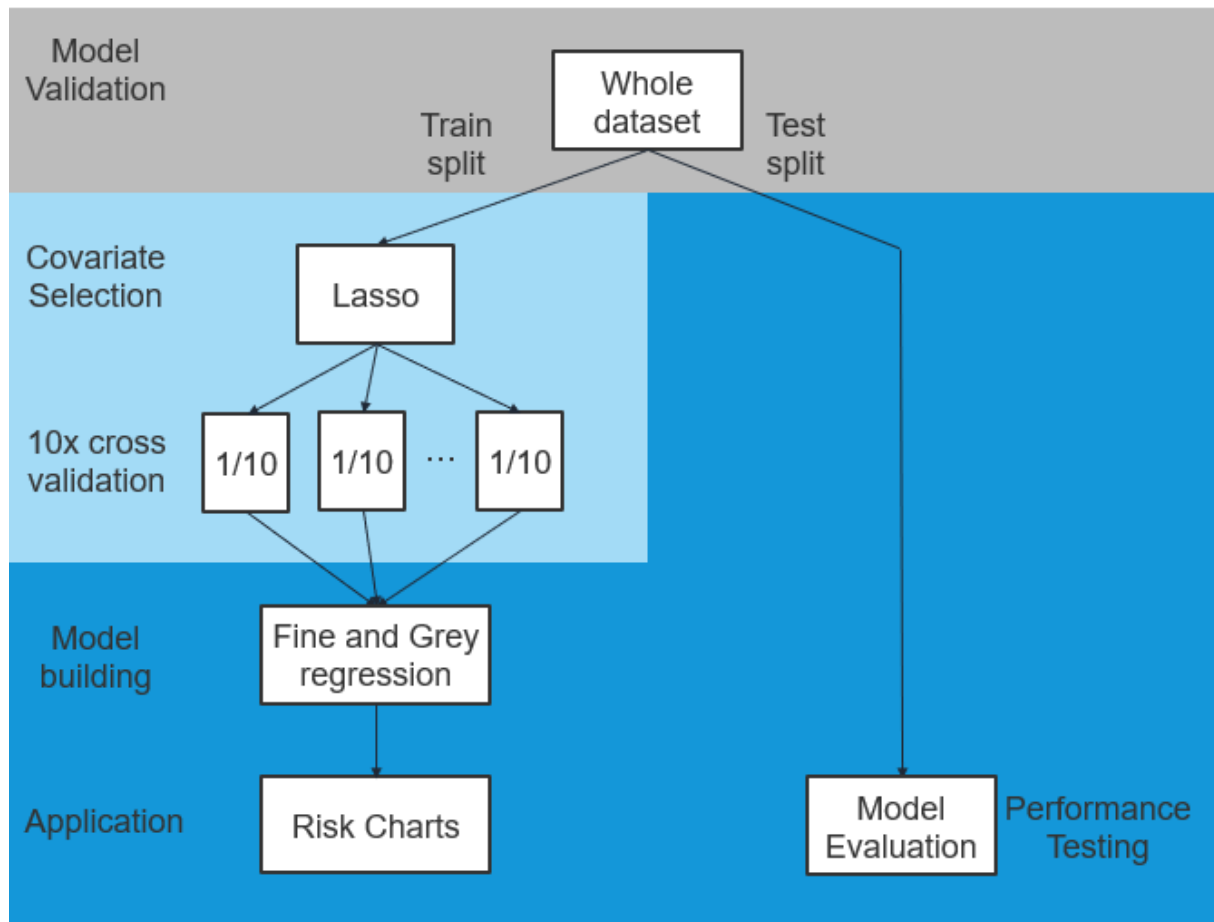


Figure 1. Schematic showing the analysis flow and splitting of data.

Preventing major information leakage from the model building process to the model testing is an important part in the creation of most prediction models. In this study, data were split using an 80:20 train-test split as displayed in Figure 1. The covariate selection and the model evaluation were performed on the training and testing splits, respectively. Since the parameters were not tuned on the testing data, the model evaluation provides a picture of how the model performs on unseen data within the UKB.

Statistics

Survival analysis

Survival analysis is the field of statistics used to analyse time-to-event data. It is a type of data where individuals are followed over time and certain events are recorded during follow-up such as a patient going into remission, dying or receiving a dementia diagnosis. The time from start of follow-up until the occurrence of the event is called follow-up time.

Censoring

When the exact follow-up time for an individual is unknown, the observation is considered censored. There are three reasons why a participant might be censored: when no event occurred before the study end, when they were lost to follow-up, or when they withdrew from the study because of an unwillingness or inability to continue participation. Non-informative censoring is caused by a random event and is thus not related to the event of interest in any way. Informative censoring occurs when the censoring is related to the event, in this context, this could occur when a person's early cognitive decline causes them to want to drop out of

the study population due to personality changes or emotional instability. Informative censoring can cause bias, leading to an underestimation of the number of people getting dementia. Unless competing risks are present, censoring is assumed to be non-informative.

Survival function

The survival function $S(t)$ describes the probability that a person survives longer than some specified time t :

$$S(t) = P(T > t) \quad 1$$

$S(t)$ is the probability of survival at time t , T is the time at which the event occurs, and t is the current time.

If one fourth of all individuals have had the event by time t_1 , $S(t_1) = 0.75$. The individuals who did not have an event before t_1 are said to be in the Risk Set, meaning that they are still able to have the event. If an individual is censored, they are removed from the risk set because no information is available about them after that time point. However, after censoring each event will have a higher impact on the survival probability since the risk set is smaller and since those individuals are assumed to represent those who were censored. For example, if half of the individuals in the risk set are censored the survival probability impact of a single event is doubled compared to if none were censored.

Hazard function

The hazard function gives the instantaneous risk that an event occurs per unit time, mathematically:

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T < t + \Delta t | T \geq t)}{\Delta t} \quad 2$$

Where $h(t)$ is the instantaneous risk, T is the time at which the event occurs, and t is the current time.

The relationship between a survival function and a hazard function is:

$$h(t) = -\frac{S'(t)}{S(t)} \quad 3$$

Which tells us that we can derive the hazard function from the survival function and vice versa. In contrast to the survival function, which gives the probability of survival, the hazard function gives the rate of failure.

Cox regression

The Cox proportional hazards model is a semi-parametric mathematical model used for survival data analysis. It uses partial likelihood optimization to relate a set of covariates to a hazard function by optimizing beta terms for each covariate. The nonparametric part of the model refers to the baseline hazard which is left unspecified. It can be determined by setting all covariates to their reference state, usually zero or the mean when applicable. The effect of the covariates is parametric since we are estimating beta parameters. In the parametric portion we are assuming that the hazards are proportional over time, and that continuous covariates are linear when related to the log hazard. In other words, interpretations from the model are dependent on the parametric portion being formulated in a way that fits the data to a sufficient degree. When selecting a statistical model, it is useful to pick the model with the smallest number of assumptions, and that provides only the interpretative space that one is interested in. In this context, we are interested in the effect of the different covariates on

dementia diagnosis. The Cox model estimates these effects by leaving the baseline hazard unspecified and estimating parameters for the covariates. (Cox, 1972; Kleinbaum & Klein, n.d.) The model looks like this:

$$h(t|\mathbf{X}) = h_0(t)e^{\sum_{i=1}^p \beta_i \mathbf{X}_i} \quad 4$$

Where $h(t|\mathbf{X})$ is the hazard function given a set of covariates, $h_0(t)$ is the baseline hazard, β_i is the coefficient for covariate i , and $\mathbf{X}_i$ is the value for covariate i .

Hazard ratios of covariates can be calculated, giving the multiplicative effect caused by the difference in the covariate (Cox, 1972; Kleinbaum & Klein, n.d.) For example, a hazard ratio of 2 would indicate a doubled risk of having the event by having diabetes compared to not having diabetes.

The proportional hazards assumption

Schoenfeld residuals can be used to assess the proportional hazards assumption over time for the Cox model. First a Cox model is fitted to obtain β coefficient estimates. Then, for each event time, the difference is calculated between the observed covariate value for the individual experiencing the event and the risk-weighted average (Equation 5) of the covariate for all individuals in the risk set. This difference is the Schoenfeld residual for that event. The formula for the risk-weighted average is as follows:

$$E(x|R(t)) = \frac{\sum_{j \in R(t)} x_j e^{\beta x_j}}{\sum_{j \in R(t)} e^{\beta x_j}} \quad 5$$

$E(x|R(t))$ is the expected value for the coefficient, or the risk-weighted average for the coefficient, given a risk set at that point in time t . The risk weight β comes from the initially fitted Cox model, for covariate x and individual j in the risk set. These Schoenfeld residuals are plotted over time, and a line is fitted through the values to visualize the function: $\beta(t) - \beta$, where deviations from a horizontal line would indicate non-proportional hazards. A score-based test, for example, a chi-squared test is used to assess whether there is an association between the residuals and time, and the resulting p -value is used to evaluate the null hypothesis. One weakness of the Schoenfeld test can be that with a lot of data the test can be too sensitive due to how p -values are calculated. (Schoenfeld, 1982; Therneau, 2001)

The proportional hazards assumption can also be visualized by plotting the log-log survival curve against the follow-up time. The hazard function in Equation 4 relates to the survival function:

$$S(t, \mathbf{X}) = [S_0(t)] e^{-\sum_{j=1}^p \beta_j \mathbf{X}_j}$$

Taking the first log results in:

$$\ln S(t, \mathbf{X}) = e^{-\sum_{j=1}^p \beta_j \mathbf{X}_j} * \ln S_0(t)$$

Since the survival function is a probability, the log will be negative, and since we need to take the log again, we multiply by -1 before the next log. And taking the second log and simplifying:

$$\begin{aligned} \ln[-\ln S(t, \mathbf{X})] &= \ln[-e^{-\sum_{j=1}^p \beta_j \mathbf{X}_j} * \ln S_0(t)] \\ \ln[-\ln S(t, \mathbf{X})] &= \sum_{j=1}^p \beta_j \mathbf{X}_j + \ln[-\ln S_0(t)] \end{aligned}$$

$$\ln[-\ln S(t, \mathbf{X})] - \ln[-\ln S_0(t)] = \sum_{j=1}^p \beta_j X_j \quad 6$$

In Equation 6 we can see that the difference between the two survival functions $S(t, \mathbf{X})$ and $S_0(t)$ is not dependent on time, meaning that these two functions should be parallel over time for the assumption to be verified. The common practice is that they need to be approximately parallel and at least not crossing for hazards to be considered proportional over time. (Kleinbaum & Klein, n.d.)

Linearity of continuous covariates

The Cox model assumes that the effect of continuous covariates on the log hazard is linear. Martingale residuals can be used for qualitative assessment of the continuous covariate's effect on the log hazard. For each individual, a martingale residual is given, and it is the difference between the number of events and the expected number of events predicted by the specified model. If these residuals are randomly distributed over the continuous covariate values, the assumption is verified according to the test. For example, a curve that goes downwards and then flattens would indicate that fewer events than expected occurred for higher covariate values and a log transformation of the covariate could compensate for that trend. (Therneau et al., 1990; Therneau & Grambsch, 2000)

The assumption of linearity can also be verified through spline plots where the continuous covariates are transformed, allowing them to curve if doing so results in a better fit. The log hazard as a function of the covariate can then be visualized for different transformations to see whether a linear relationship is kept across the transformations. If the linear relationship is kept, the assumption is verified.

Interaction analysis

Cox regression estimates the cause specific hazard for each included covariate. Interacting effects can be assessed to get a picture of whether two covariates have a greater (or smaller) effect depending on their combined state. For example, if having both hypertension and heart disease has a larger effect than either covariate alone in isolation it would mean that they have a positive interaction effect. An exhaustive pairwise test was performed, and the number of required tests can be calculated as:

$$N = p(p - 1)/2 \quad 7$$

Where N is the number of tests, and p is the number of covariates. This is computationally feasible considering the number of covariates intended for inclusion in the analysis. A likelihood ratio test was used to test the significance of the impact of different interactions compared to a basic model without interactions (Cox, 1972). In order to correct for multiple testing, the Bonferroni correction was used (Holm, 1979). When performing multiple tests in the same analysis, it is required to adjust the significance level such that the number of tests performed (Equation 7) is accounted for to prevent false positives. The significance level 0.05 was divided by the number of tests to obtain the Bonferroni adjusted significance level.

Covariate selection

Cox regression maximizes a partial likelihood formula.

$$\ell(\beta) = \sum_{i:\delta_i=1} \left(\beta^T x_i - \log \left(\sum_{j \in R(t_i)} e^{\beta^T x_j} \right) \right) \quad 8$$

Where $\ell(\beta)$ is the log likelihood given a vector of β coefficients. δ_i is the event indicator which assumes the value 1 if subject i experiences the event. x_i is the covariate vector for an individual who experienced the event. x_j is the covariate vector for all individuals in the risk set $R(t_i)$ immediately prior to the event time t_i .

In this formula the beta terms are adjusted to fit the data in the best possible fashion. Using this we will get a picture of the strength of the impact of the different covariates. It is, however, also useful to get a picture of the level of importance of each covariate. This can be done to exclude certain covariates, which in turn allows for the creation of a simpler and thus a more interpretable model. Since the goal is for this model to be used in a preventive setting, where interpretation is key, this is of high importance. Lasso regression introduces a penalty term into the optimization algorithm of Equation 8.

$$\ell_{LASSO}(\beta) = \ell(\beta) - \lambda \sum_{k=1}^p |\beta_k| \quad 9$$

Where $\ell_{LASSO}(\beta)$ is the lasso log likelihood at the value of the coefficient vector β with lasso adjustments. λ is the regularization term. p is the number of predictors. $\sum_{k=1}^p |\beta_k|$ is the lasso penalty whose impact is controlled by λ .

The goal of the optimization is to obtain the largest possible value in the log likelihood. The penalty term rewards the optimization algorithm for decreasing the value of the beta coefficients as seen in Equation 9, and as the regularization term λ increases, the beta coefficients decrease. The coefficients that are first to reach zero can be deemed to be of least importance, since they were the first to be excluded by the optimization algorithm. (Tibshirani, 1996). Prior to fitting the Cox lasso model, the covariate values were scaled by 'glmnet' to ensure that the penalty is applied evenly across the covariates, otherwise the coefficient penalties might be skewed.

To get a picture of the loss in log partial likelihood from the lasso regression, and to pick out optimal penalty values for a minimal and maximal subset, cross-validation was used. This means that the data is partitioned and a Cox lasso regression is run for each partition. Using the partial log-likelihood values for each run a visualisation can be created showing the deviance in the partial log-likelihood for each value of the penalty term.

Fine and Gray-regression

A competing risk is an event type that prevents the event of interest from occurring. For example, a person who is dead is unable to be diagnosed with dementia and should therefore not be interpreted as a censored event. In the situation where there are competing risks, Fine-Gray risk regression can be utilized.

A cumulative incidence curve (CIC) gives the proportion of individuals who have experienced the event after a time t . The Fine and Gray model is used to estimate the subdistribution hazard from which the cumulative incidence function is derived and visualized with a CIC. The main difference to the Cox model is that the individuals who experience a competing event are kept, with an applied weight, in a modified risk set; instead of being removed from the analysis entirely due to any other censoring event. (Fine & Gray, 1999; Kleinbaum & Klein, n.d.)

Risk Charts

Risk charts are widely used in the medical field for risk assessment and prevention of diseases. They provide estimated risk over different combinations of covariate values, such as

age brackets, genotypes, sex, diabetes status, etc. Using a fitted model, every individual’s risk score is calculated and converted into an absolute risk score based on the period over which the risk is projected across (e.g., 10 years). Covariates to be included in the model are chosen carefully, since too many covariates can make a risk chart messy and hard to interpret. For the chosen covariates, a risk is calculated for every combination (Cox, 1972). By identifying an individual’s combination of the covariates included in the risk chart, their risk of developing dementia over a set amount of time can be estimated with ease.

Performance evaluation

The Receiver Operating Characteristic (ROC) curve is a classical method for evaluation of binary classifiers. Together with the Area Under the Curve (AUC) curves, they can provide an unbiased overview of performance also for ranges of follow-up time or age brackets. The ROC curve plots sensitivity (True Positive Rate) against 1 - specificity (false-positive rate) over a range of threshold values. The area under the curve measures classifier performance where an AUC of 0.5 is completely random. ROC curves may then be created over a range of age brackets where the AUC is calculated for the ROC curves yielding AUC curves.

Software

Data formatting and analysis was done on the UK Biobank Cloud RStudio with R version 4.5.3. The ‘tidyverse’ was used mainly for data formatting with some exceptions. Standard survival analysis models were made using the ‘survival’ package. Fine-Gray regression was done with the ‘riskRegression’ package. Lasso regression was performed using the ‘glmnet’ package. All code and a renv lockfile can be found at https://github.com/Sladderdask/dementia_risk.

Results

UKB characteristics

Individuals of non-Caucasian descent or with missing values in any of the investigated categories were excluded from the analysis. This resulted in a total study population exceeding 400,000 individuals after filtration. Unfortunately, the results presented in this section are limited due to the UKB shutdown. The following table is based on data from the master’s thesis of Minois Genin where the UKB was used as the cohort for dementia risk prediction.

Table 3. Characteristics of the included individuals in the UK Biobank from the master's thesis of Mi-nois Genin.

No. of risk alleles	Female				Male			
	47 – 68	68 – 71	71 – 75	75 – 99	47 – 68	68 – 71	71 – 75	75 – 99
No. of individuals	61173	55364	77985	69493	51500	46463	65494	59343
Age (years)	56.7	56.8	56.9	57.0	57.1	57.2	57.3	57.4
Deaths (%)	6.5	6.6	6.6	6.9	11.2	11.2	11.4	11.5
T2DM (%)	7.3	7.0	6.8	6.4	12.5	11.8	11.7	11.4
Stroke (%)	0.5	0.5	0.5	0.5	1.0	1.1	0.9	0.9
Hypertension (%)	23.2	22.9	22.4	21.9	31.6	31.0	30.9	30.3
non-HDL (mmol/L)	4.3	4.3	4.3	4.3	4.2	4.2	4.2	4.2
Education (%)	63.3	63.2	63.2	63.3	68.5	68.5	68.9	68.2
Alcohol (%)	35.4	36.3	37.1	37.6	50.3	50.8	51.9	52.7
APOE genotype (%)								
ϵ 22	0.7	0.6	0.7	0.6	0.6	0.6	0.7	0.6
ϵ 32	12.3	12.3	12.0	12.3	12.2	12.2	12.3	12.2
ϵ 33	58.3	58.6	59.2	58.6	58.7	58.6	58.8	58.8
ϵ 42	2.6	2.5	2.5	2.4	2.7	2.6	2.4	2.4
ϵ 43	23.8	23.6	23.3	23.6	23.4	23.6	23.5	23.5
ϵ 44	2.4	2.3	2.4	2.4	2.4	2.4	2.3	2.5
Dementia subtype (%)								
Alzheimer	0.6	0.7	0.8	1.2	0.6	0.7	1.0	1.3
Vascular	0.3	0.3	0.3	0.5	0.5	0.5	0.5	0.7
Vascular related	0.8	1.0	1.1	1.4	1.3	1.3	1.5	1.8
All dementia	1.2	1.4	1.5	2.1	1.6	1.7	2.0	2.5

Proportional hazards assumption evaluation

Schoenfeld residuals tests can be used for verification of the proportional hazards assumption made in Cox regression. This is an essential step in the creation of a Cox model to make defensible interpretations of the results. In Figures 2 and 3, these results are shown for AD and VD, respectively.

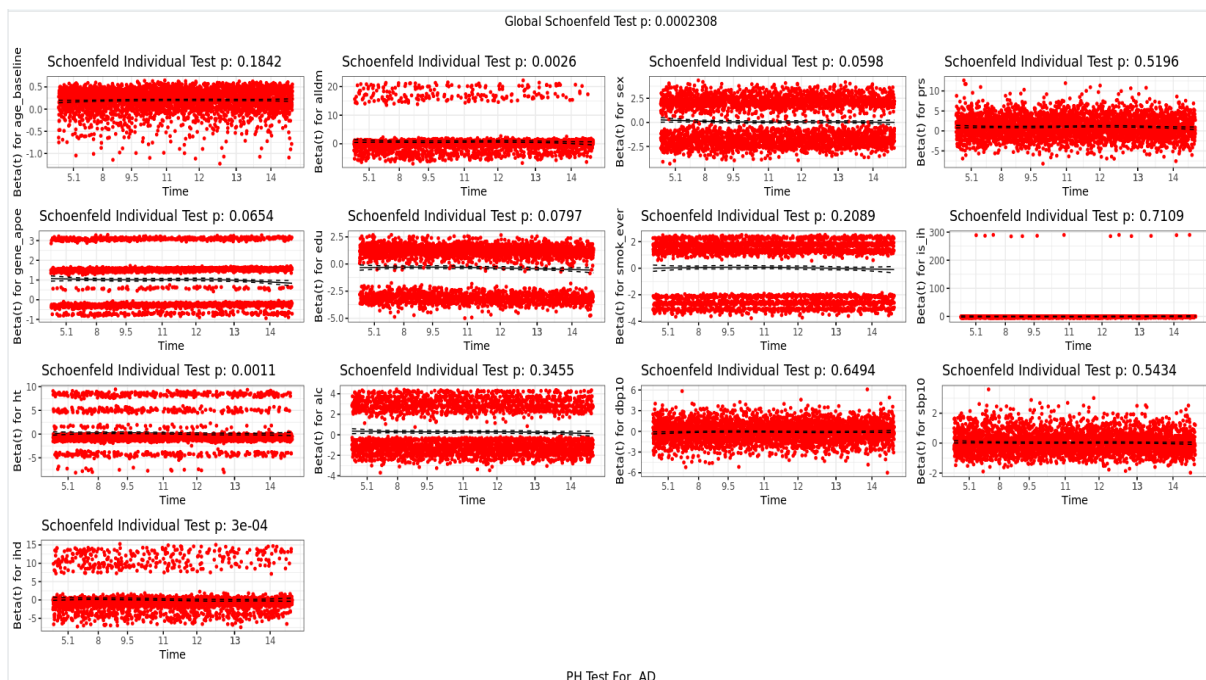


Figure 2. Scaled Schoenfeld Residual tests for AD. Solid black lines represent smoothed estimates of time-varying regression beta coefficients, and dashed lines represent confidence intervals. Each red dot is a Schoenfeld residual from which the smoothed line is estimated. P-values indicate the significance level of the Schoenfeld residual test for the proportional hazards assumption, where the null hypothesis is that the beta coefficients are constant over time.

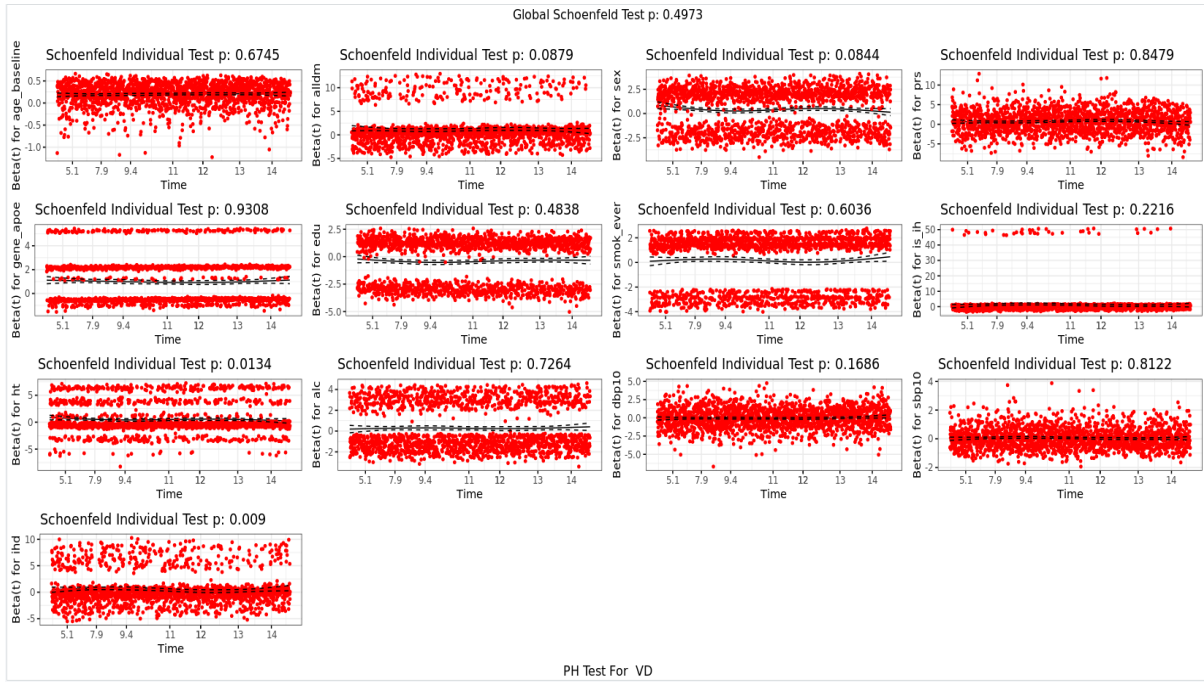


Figure 3. Scaled Schoenfeld Residual tests for VD. Solid black lines represent smoothed estimates of time-varying regression beta coefficients, and dashed lines represent confidence intervals. Each red dot is a Schoenfeld residual from which the smoothed line is estimated. P-values indicate the significance level of the Schoenfeld residual test for the proportional hazards assumption, where the null hypothesis is that the beta coefficients are constant over time.

In Figure 2, using AD as the endpoint, the hazards are not proportional for diabetes (alldm), hypertension and ischemic heart disease (ihd). In Figure 3, the hazards are proportional for hypertension and ischemic heart disease (ihd). Due to the high sensitivity of the Schoenfeld residuals test in combination with the large dataset, it is useful to verify the significant results. Log-log plots provide a more visual assessment of the proportionality of hazards. Unfortunately, due to the UKB shutdown, the covariate types do not align exactly with the previous results.

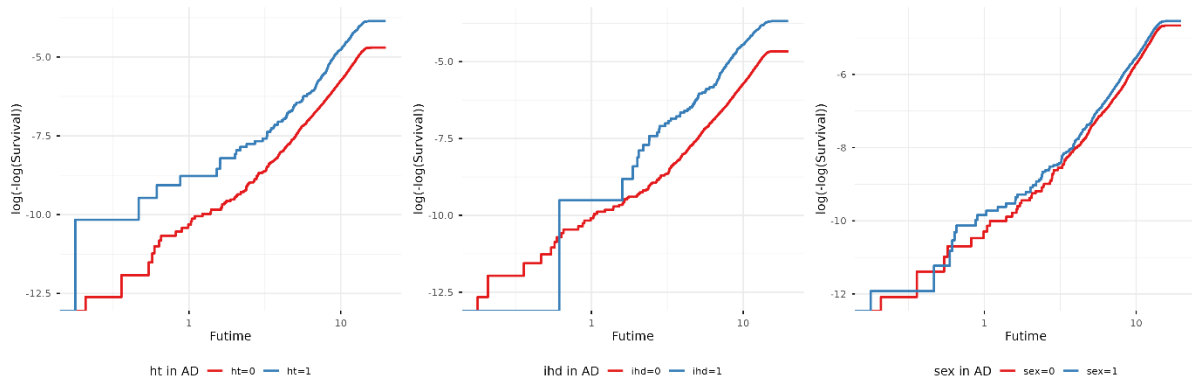


Figure 4. Log-log plots for proportional hazards assessment for AD. The x- and y-axes show follow-up time (Futime) and $\log(-\log(\text{survival}))$ time, respectively. For hypertension (ht) and ischemic heart disease (ihd) covariates, 1 indicates presence of disease at baseline. For sex: 0 stands for female and 1 indicates male. Different covariate values resulting in approximately parallel curves are consistent with proportional hazards over follow-up time.

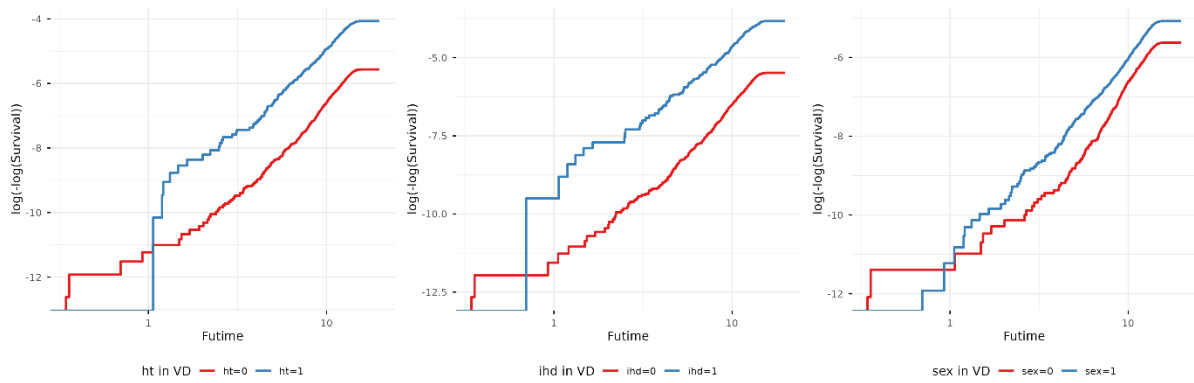


Figure 5. Log-log plots for proportional hazards assessment for VD. The x- and y-axes show follow-up time (Fuptime) and $\log(-\log(\text{Survival}))$ time, respectively. For hypertension (ht) and ischemic heart disease (ihd) covariates, 1 indicates presence of disease at baseline. For sex: 0 stands for female and 1 indicates male. Different covariate values resulting in approximately parallel curves are consistent with proportional hazards over follow-up time.

Due to the low number of failures during early follow-up, there is a strong stepwise tendency in that region and occasionally this causes the curves to cross in Figures 4-5. It is common practice not to allow crossing of the lines. However, due to the low number of data points this can be disregarded. Thus, the interpretation is that the hazards are proportional for these covariates.

Log-hazard linearity assumption validation

The assumption that covariates are linearly related to the log hazard was assessed. Three Cox models were fit for each covariate to get a picture of the change in the log hazard for different degrees of freedom. If the higher orders deviated drastically from the lower-order models, it would indicate that the covariate is not linearly related to the log hazard. Unfortunately, only spline plots were created before the UKB shutdown. Otherwise, martingale residuals would be able to provide complementary evidence for validation.



Figure 6. Spline plots of Cox regression covariates with one (red), two (green) and three (blue) degrees of freedom for continuous covariates for AD. The density distribution of the covariate is shown in the steel blue area at the bottom of the plot. The curves show the mean-centred relationship between log hazard ratio and covariate, and the shaded areas are 95% confidence intervals of these curves. The Cox model assumes a linear relationship, meaning that a straight-line relationship supports the assumption.



Figure 7. Spline plots of Cox regression covariates with one (red), two (green) and three (blue) degrees of freedom for continuous covariates for VD. The density distribution of the covariate is shown in the steel blue area at the bottom of the plot. The curves show the mean-centred relationship between log hazard ratio and covariate, and the shaded areas are 95% confidence intervals of these curves. The Cox model assumes a linear relationship, meaning that a straight-line relationship supports the assumption.

In Figures 6 and 7, we can see that the assumption is linear for both AD and VD, because the line is roughly straight with narrow confidence bands for fits with one, two and three degrees of freedom. This means that no transformations of covariates need to be made in the model definition for the assumption to hold. The blood pressure measurements dbp10 and sbp10 show varying degrees of curvature for both AD and VD as seen in Figures 6 and 7, respectively. However, the curves generally show a linear trend in the more densely populated region as well as very wide confidence intervals in the sparsely populated high and low blood pressure regions. This argues for either keeping the variable as it is for ease of interpretation or partitioning the covariate into regions where different interpretations are made for low, medium and high blood pressure.

Interaction analysis

An exhaustive pairwise search for interactive effects was performed and no significant or meaningful interactions were identified. No numbers or tables were recorded before the UKB shutdown.

Effect of covariates in Cox-models

When the model assumptions had been adequately validated, forest plots were created to visualize the hazard ratios. For each covariate, four different Cox models were fitted: the first model was fitted with sex, age and covariate; the second model with sex, age, genetic covariates and covariate; the third model with all covariates; and the fourth model with all covariates and genetics. The benefit of fitting several models like this is that they provide a picture of how some covariates might be able to explain the effects of others. For example, if the effect of a stroke shrinks when hypertension is included it might indicate that both covariates explain the same underlying biological phenomenon. The hazard ratios for each of these models and covariates were visualized in a forest plot where the effects of the covariates can be compared in the different covariate environments and among one another. 95% confidence intervals were used to assess the significance of the findings. The hazard ratios for covariates in the four different model configurations for AD are presented in Figure 8 and for VD in Figure 9. Since VD was found to have sufficient statistical power to be interpretable, as seen in Figure 9, the VRD results are shown in the appendix in Figures A1-A6.

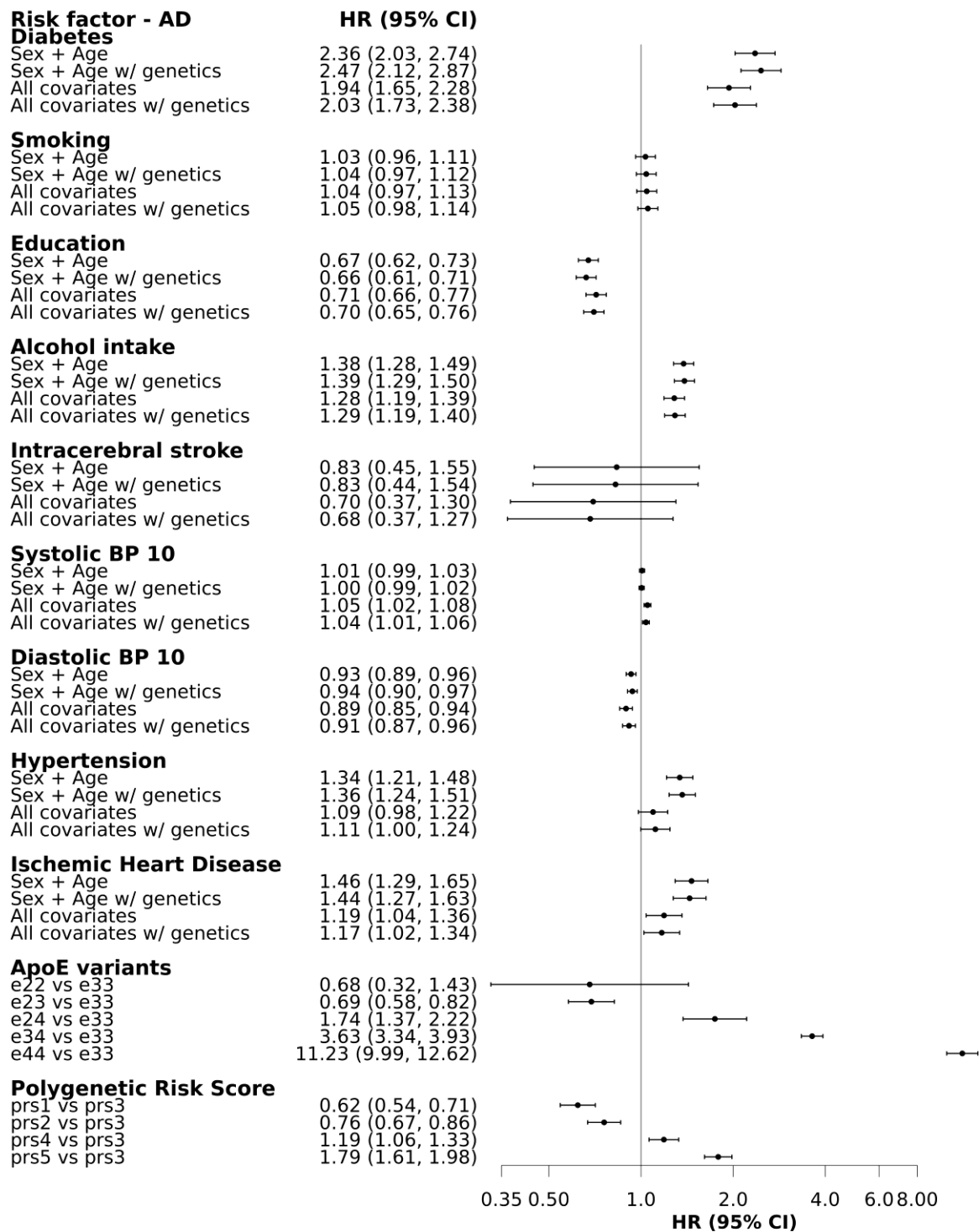


Figure 8. Forest plot showing hazard ratios for Cox models for AD. Four different models were fitted for each covariate: first with sex, age and covariate; second with age, sex, genetics and covariate; third with all covariates excluding genetics; fourth with all covariates including genetics. The mean hazard ratio (HR) is shown with a 95% confidence interval. The HR for blood pressure is shown in steps of 10 mmHg.

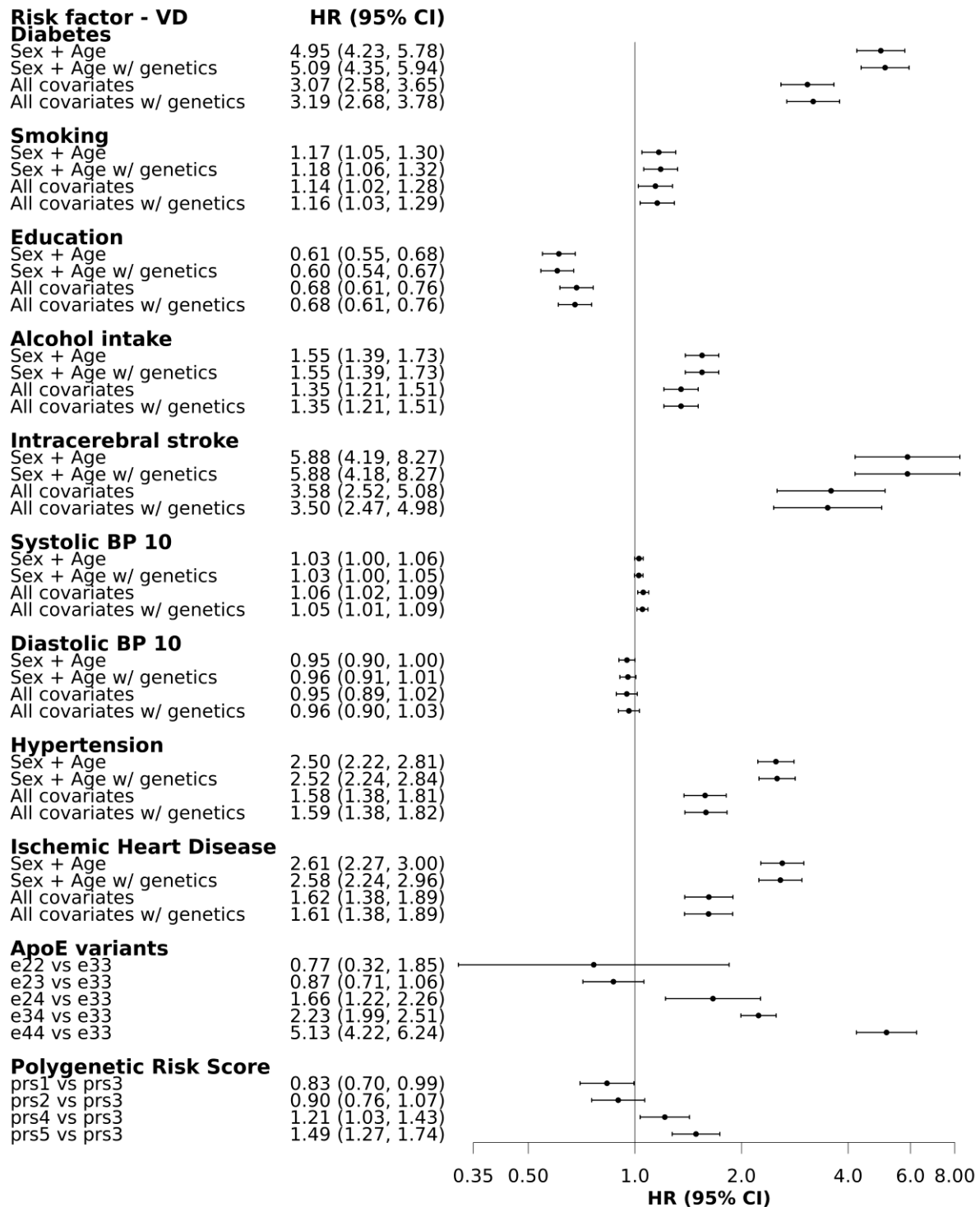


Figure 9. Forest plot showing hazard ratios for Cox models for VD. Four different models were fitted for each covariate: first with sex, age and covariate; second with age, sex, genetics and covariate; third with all covariates excluding genetics; fourth with all covariates including genetics. The mean hazard ratio (HR) is shown with a 95% confidence interval. The HR for blood pressure is shown in steps of 10 mmHg.

For AD the APOE44 variant has by far the highest impact with a hazard ratio of 11.23. High education and the APOE22 and APOE23 variants are associated with protective hazard ratios of 0.70, 0.68, and 0.69 respectively. A very wide confidence interval as well as a non-significant protective effect was seen for stroke. This might be due to the fact that in the presence of a stroke, a patient is more likely to receive a mixed dementia or VD diagnosis rather than AD,

giving the impression that it is protective against AD diagnosis. For VD all vascular risk factors: diabetes, intracranial stroke, blood pressure, hypertension, and heart disease have greater hazard ratios compared with AD, which is expected for the disease etiology. Genetic risk factors have a slightly smaller but still strong impact on VD. The effect of diabetes, hypertension and ischemic heart disease is reduced significantly after introducing other covariates in VD, meaning that the effect of diabetes can be explained by other covariates.

Selection of covariates

After getting a first glance into the hazard ratios of the different risk factors, a lasso regression was performed to get a picture of the importance of the different risk factors on AD and VD. The tests were separated into inherent and lifestyle-based risk factors.

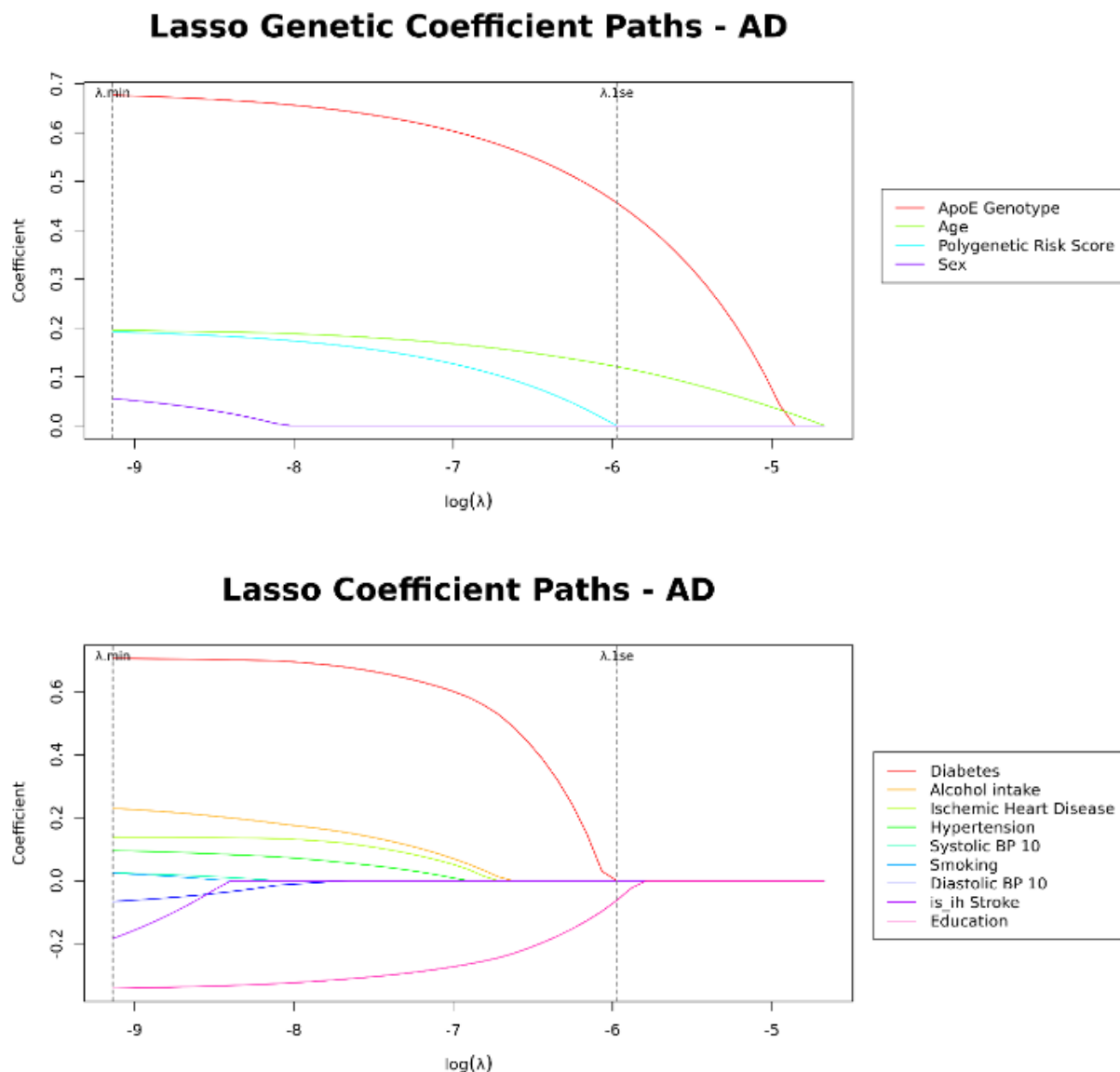


Figure 10. Upper and lower figures show Lasso regression for inherent and modifiable risk factors respectively for AD. The figures show how the coefficients (β , y-axis) vary over different values for the penalty term (λ , x-axis). The maximum and minimum defensible penalty values are indicated with $\lambda.min$ and $\lambda.1se$ respectively using dotted lines. Covariates are colour coded. A covariate's coefficient which reaches zero at a lower penalty value is deemed less important and is more likely to be excluded.

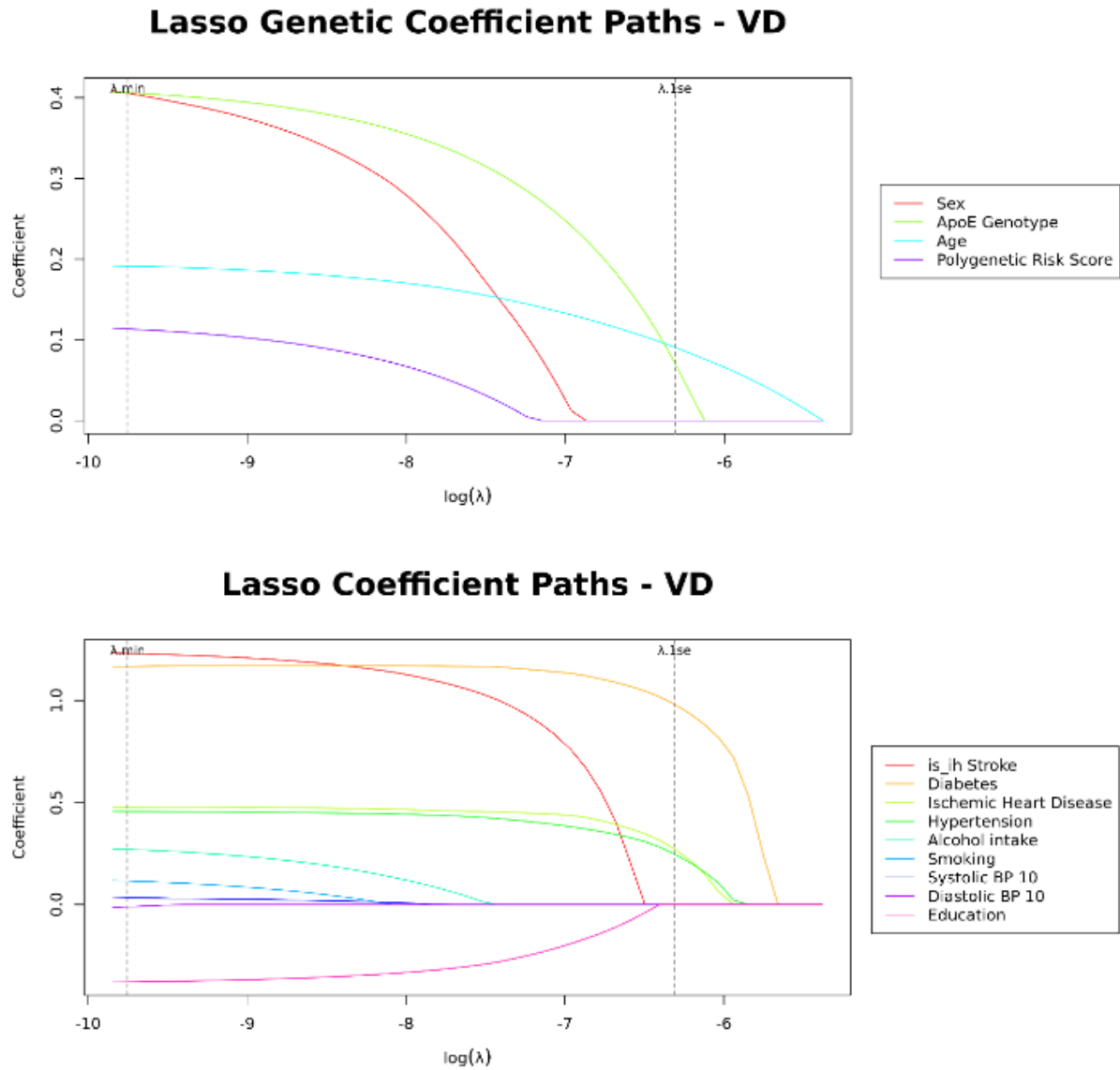


Figure 11. Upper and lower figures show Lasso regression for inherent and modifiable risk factors respectively for VD. The figures show how the coefficients (β , y-axis) vary over different values for the penalty term (λ , x-axis). The maximum and minimum defensible penalty values are indicated with λ_{min} and λ_{1se} respectively using dotted lines. Covariates are colour coded. A covariate's coefficient which reaches zero at a lower penalty value is deemed less important and is more likely to be excluded.

To get a picture of the loss in partial log likelihood, lasso cross-validation was performed over different values of the penalty value. This provides information for selecting a suitable number of covariates to be included in a minimal model. This is the test from which the values for λ_{min} and λ_{1se} in Figures 10 and 11 were calculated. In Figures 12 and 13, this curve is visualized.

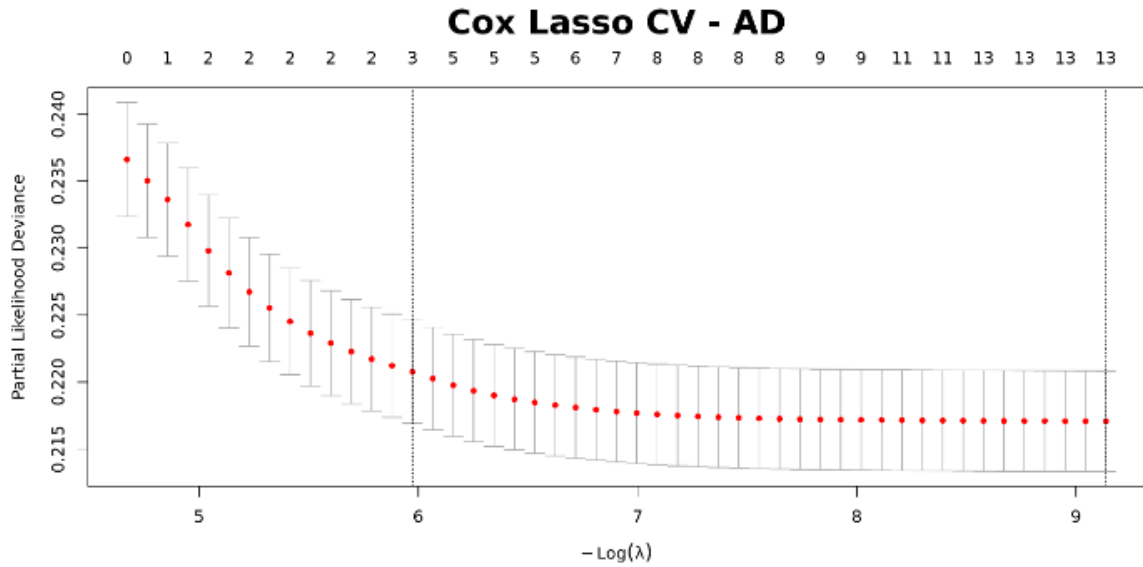


Figure 12. Lasso cross-validation of Cox models for AD. The partial likelihood deviance (y-axis) as a function of penalty values (λ , x-axis) over a 10-fold cross-validation. The mean values of the cross-validation runs are shown in red, with corresponding 95% confidence intervals. At the top of the plot the total number of covariates included at the corresponding values of the penalty term are indicated.

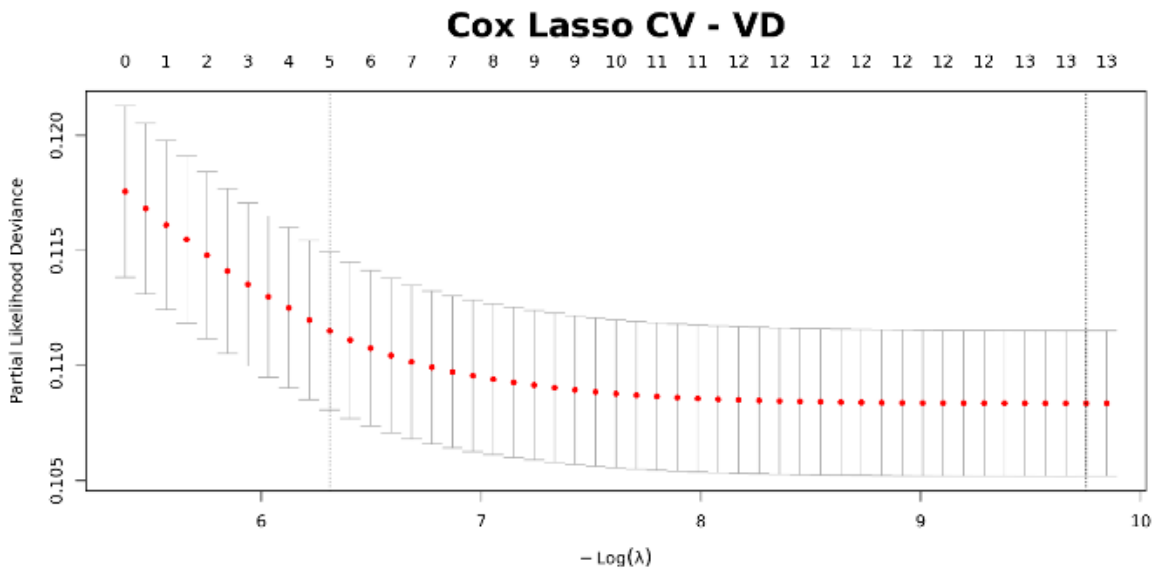


Figure 13. Lasso cross-validation of Cox models for VD. The partial likelihood deviance (y-axis) as a function of penalty values (λ , x-axis) over a 10-fold cross-validation. The mean values of the cross-validation runs are shown in red, with corresponding 95% confidence intervals. At the top of the plot the total number of covariates included at the corresponding values of the penalty term are indicated.

For AD, using the λ_{1se} cutoff from Figure 12 it is seen that the minimum model incorporates only three covariates. Furthermore, using Figure 10, we see that those covariates are age, APOE genotype and education. Also, the maximum model is using all covariates, meaning that all included covariates provide useful information for the predictions. For VD, using the same method of interpretation, five covariates: diabetes, ischemic heart disease, hypertension, APOE genotype and age are included in the minimal subset. While all covariates contribute some information, it is noted that more covariates are deemed necessary for VD in

comparison to AD. This is expected because several vascular risk factors are included in the prediction model and since they are closely tied to the nature of the disease.

Performance testing and risk charts

At the time of UKB shutdown Fine-Gray models had converged, performance testing and risk chart figure creation was still underway and therefore not completed (Nightingale, 2026).

Discussion

Using over 400,000 participants in the UKB, risk prediction models were created for AD with 3 covariates and for VD with 5 covariates. The models were developed through a rigorous exploration of model assumptions and cross-validated lasso regression for covariate selection. The significant Schoenfeld residual tests in Figures 2 and 3 were disproven by visual inspection of log–log plots in Figures 4 and 5 and linear relationships between covariates and log hazards were shown using splines in Figures 6 and 7. Having completed the model-validation steps, no alterations to the covariates such as transformations or partitioning were required. Hazard ratios of Cox models for AD and VD were presented using forest plots in Figures 8 and 9. Covariate selection was performed using Lasso regression and cross-validation, where AD was found to require only the APOE genotype, age and education, as seen in Figures 11 and 12. VD was found to require only the APOE genotype, age, diabetes, ischemic heart disease and hypertension, as seen in Figures 11 and 13. Performance testing and risk charts were not completed due to a UKB shutdown.

The results underpin the possibilities as well as the difficulties in prevention of AD and VD. Considering the preventability of AD, the genetic covariates and age show the highest effects, as seen in the forest plot in Figure 8 and lasso plots in Figure 10. For the modifiable risk factors there is a clear association for both education and alcohol intake suggesting some potential for prevention. This is in accordance with previous research and theory regarding cognitive reserve and the chemotoxic and genotoxic effects of alcohol metabolism (Zakhari, 2006). Genetic factors are estimated to explain 60–80% of the variation in Alzheimer's disease risk (Gatz et al., 2006), supporting the expectation of not finding comparatively large hazards for lifestyle-associated risk factors. The greater preventability of VD is evident both in the forest plot in Figure 9 and the lasso cross-validation in Figures 11 and 13, particularly in comparison to AD. The preventability is seen in higher hazard ratios as well as a greater number of covariates required for non-significant loss in predictive accuracy. The effect of diabetes, alcohol, hypertension and ischemic heart disease is weakened after inclusion of the other covariates, suggesting a confounding effect. However, according to the cross-validation, all covariates contribute to prediction accuracy. Furthermore, all are significant after inclusion of the others. Due to the clearer biological association between intracranial stroke and VD, one may find it strange to see that ischemic heart disease is deemed the more important predictor in the lasso regression. This may, however, be due to the rarity of strokes at baseline in the UKB, as indicated by the wide confidence intervals for stroke in Figures 8 and 9, reducing the statistical power for this covariate.

Rasmussen et al. evaluated the impact of risk factors on all dementia types with participants from the Copenhagen general population study and Copenhagen City Heart study, $N = 61\,664$ aged 20–100 years (Juul Rasmussen et al., 2020). While the study involved a smaller dataset and did not partition between disease types, some similarities and dissimilarities in their findings can be ascertained. The present report with more observations gave rise to more statistically significant observations, which is likely a result of statistical power, as well as a greater distinction between endpoints because they are distinguished by different diagnoses.

Combining diseases into a single endpoint might dilute observations since risk factors differ between disease subtypes. Prospects could involve further partitioning into AD and VD subtypes. Such analyses might reveal subtype-specific risk factors that are hidden at the current level of disease categorization.

Akin to the present work, the master’s thesis of Minois-Genin (Minois-Genin, 2025) explored risk factors using the UKB cohort. This study has delved deeper into the assumptions and specifications of the model, providing a more detailed understanding of the overall behaviour of the model. The stronger evidence for the absence of model violations increases the legitimacy of the estimated hazard ratios. In contrast with the results of Minois-Genin, alcohol was found to have a positive association with AD and VD which is more in accordance with current biological understanding, which may be the cause of a greater focus on model specification in the present study. This finding may contribute to a clearer understanding concerning the effect of alcohol. Whereas Minois-Genin relied primarily on lasso regression alone, the current study was extended by employing cross-validation of the Cox lasso regression. The additional insight into the loss of accuracy per included covariate gives more legitimacy for the selection of the number of covariates to include in a dementia prediction model.

This work has contributed to dementia risk-prediction research through a detailed analysis of the model assumptions, effects and importance. However, future research could aim to fill some gaps. The unforeseen and sudden UKB shutdown prevented finalization of the work. This limited the work overall in numerous ways such as the inability to rerun analyses for adjustment of figures during the writing process, leading to decreased flexibility, interpretability and clarity during presentation of the results. It also caused the loss of results such as summary tables, martingale residuals, interaction analysis, performance assessment and risk charts. This led to overall incompleteness of the project. Additionally, due to the Caucasian-dominated composition of the UKB, other ancestries were removed from the cohort to reduce variance caused by genetic differences. This may reduce prediction accuracy for non-Caucasians, reducing the utility of the results overall.

The age distribution and follow-up time of the cohort weaken the potency of the fitted models. A large portion of the cohort is still in their 60s and 70s and thus still have a low likelihood of having developed dementia. This limits the reliability of the results because there are many censored subjects, increasing the sensitivity of the results to the model assumptions. As the biobank matures, and more individuals develop dementia or die, the potency of the produced models will increase. The assumption of independent survival times between observations has not been tested. Due to the recruitment strategy, which relied on recruitment centres within a geographical radius, some individuals are genetically related, share socioeconomic conditions and might have similar lifestyles. This assumption was not investigated given the size of the cohort, which is expected to limit the impact of violations of this assumption. However, evaluating the extent of the violations would have been useful to better understand its potential effect. Disease covariates were defined at baseline and not updated during follow-up. By incorporating updating covariates into the model, a more efficient use of the data could have been attained, which may more closely reflect the real association between the diseases and endpoints.

For most observational studies the causal nature of the findings comes into question. To minimize this effect a thorough review of the literature on the modifiable risk factors was performed to provide a biological motivation for any associations. The plan was also to exclude type 2 diabetes for AD due to the causal link recently coming into question (Han et al., 2025a; Litkowski et al., 2023).

Conclusion

A cause-specific survival model was created for AD and VD with rigorous testing of statistical assumptions. These findings have given greater insight into the behaviour and importance of risk factors on AD and VD, one step forward toward a greater understanding of dementia prevention. A sudden shutdown of the UKB platform prevented the creation of several important figures and analyses; however, much was still able to be distilled from the completed work.

Bioethical impact statement

This work has involved analysis of previously collected, de-identified human data from the UKB, including genetic, lifestyle and health information. The UKB has research ethics approval from North West – Haydock Research Ethics Committee (REC reference 16/NW/0274) and all participants have provided informed consent (“Ethics,” 2025). This study was part of an approved UKB research project with project ID 66414 (“Personalized Prevention of Dementia - Integrating Lifestyle, Biomarkers and Genomics.,” 2025).

Societal impact statement

The societal burden of dementia is expected to increase as demographics change and the mean age of the population increases (Collaborators, 2022). Since medical treatment options are limited (Birks & Harvey, 2018; Cohen et al., 2023; Matsunaga et al., 2015), attention has shifted toward preventive approaches, with studies reporting that up to 40 percent of cases can be delayed or prevented through management of modifiable risk factors (Livingston et al., 2024). This study used statistical analysis of these risk factors to gain a deeper understanding of their consistency with survival-model assumptions, relative importance and interactions, as well as producing disease-specific risk charts. The main findings include robust verification of the Cox model assumptions for the included covariates and an assessment of their hazard ratios for both AD and VD. Interaction terms were concluded not to have a significant impact. Cross-validation showed that models with only 3 covariates for AD and 5 for VD resulted in no significant decrease in predictive accuracy. These findings give greater insight into the impact and behaviour of risk factors on AD and VD which brings the field one step closer to a complete dementia prediction and prevention workflow. Early identification of high-risk individuals will pave the way for cost-effective strategies for risk factor management.

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Appendix

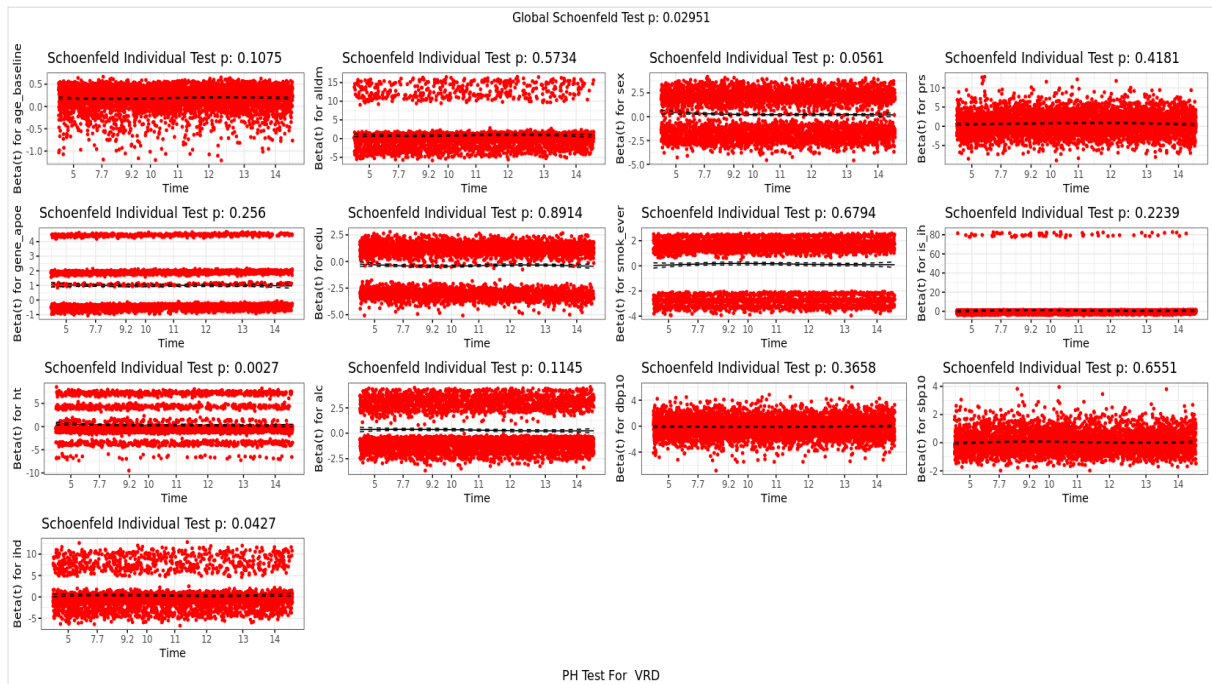


Figure A1. Scaled Schoenfeld Residual tests for VRD in a Cox regression model. Solid black lines represent smoothed estimates of time-varying regression beta coefficients, and dashed lines represent confidence intervals. Each red dot is a Schoenfeld residual from which the smoothed line is estimated. P-values indicate the significance level of the Schoenfeld residual test for the proportional hazards assumption, where the null hypothesis is that the beta coefficients are constant over time.

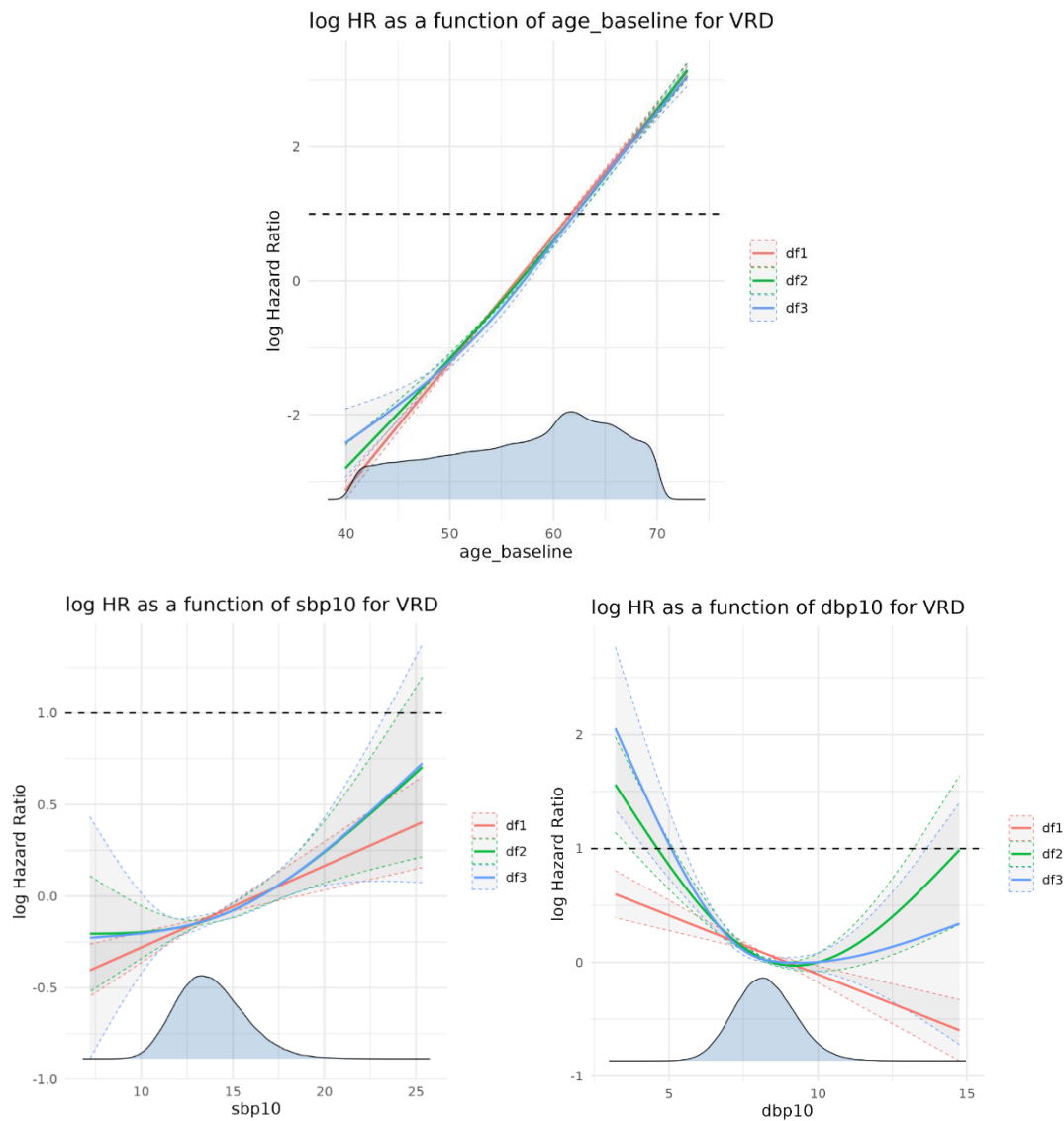


Figure A2. Spline plots of Cox regression covariates with one (red), two (green) and three (blue) degrees of freedom for continuous covariates for VRD. The density distribution of the covariate is shown in the steel blue area at the bottom of the plot. The curves show the mean-centred relationship between log hazard ratio and covariate, and the shaded areas are 95% confidence intervals of these curves. The Cox model assumes a linear relationship, meaning that a straight-line relationship supports the assumption.

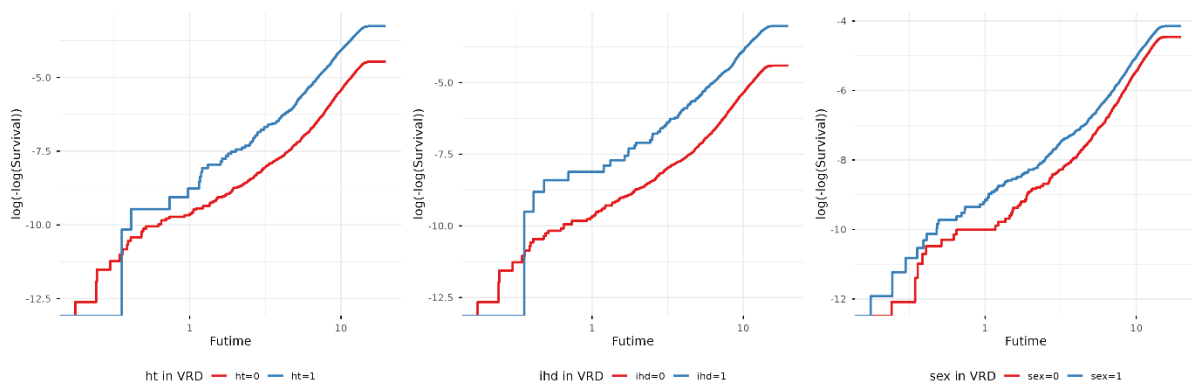


Figure A3. Log-log plots for proportional hazards assessment for VRD. The x- and y-axes show follow-up time and $\log(-\log(\text{survival}))$ time, respectively. For hypertension (ht) and ischemic heart disease (ihd) covariates, 1 indicates presence of disease at baseline. For sex: 0 stands for female and 1

indicates male. Different covariate values resulting in approximately parallel curves are consistent with proportional hazards over follow-up time.

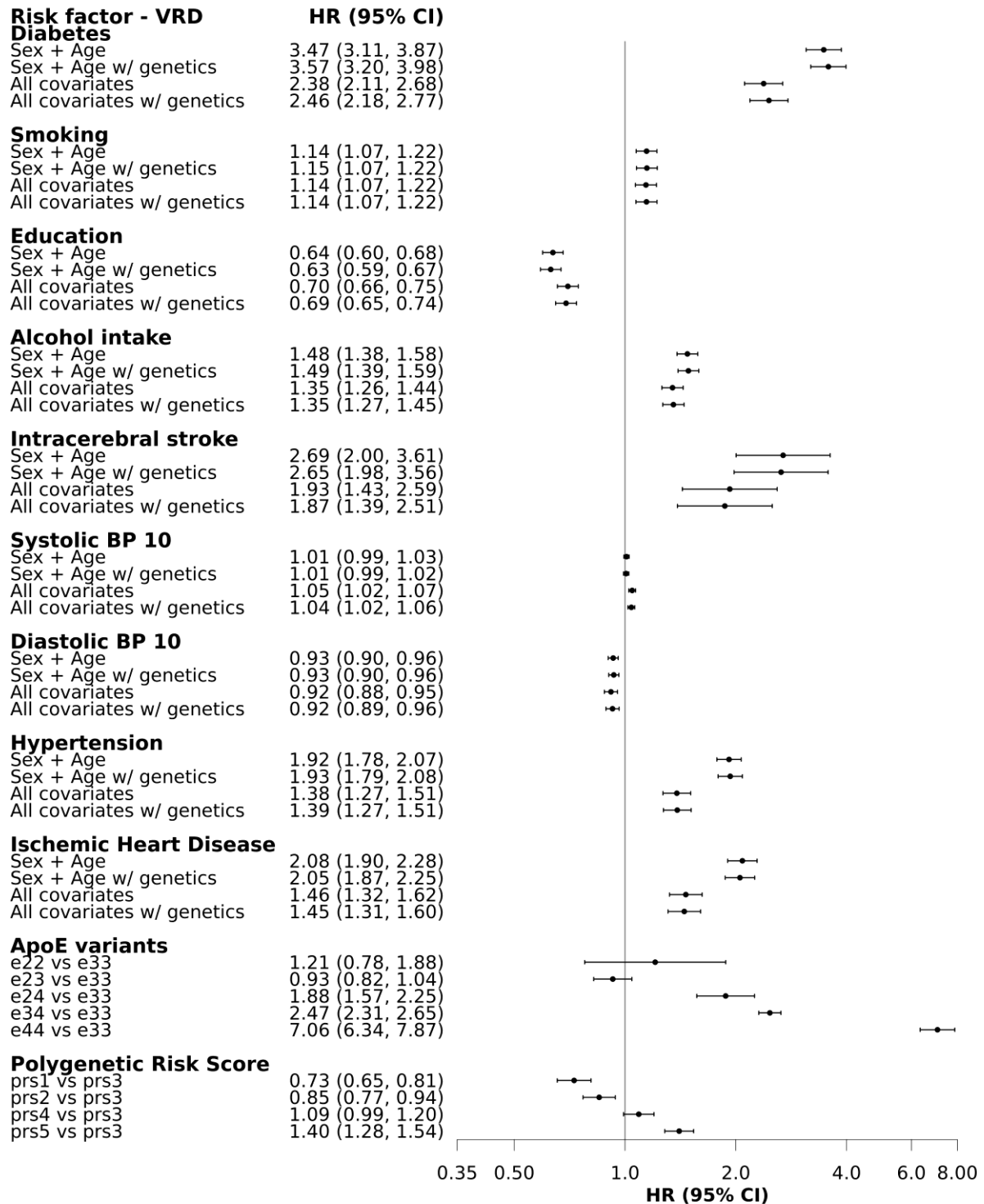


Figure A4. Forest plot showing hazard ratios for Cox models for VRD. Four different models were fitted for each covariate: first with sex, age and covariate; second with age, sex, genetics and covariate; third with all covariates excluding genetics; fourth with all covariates including genetics. The mean hazard ratio (HR) is shown with a 95% confidence interval. The HR for blood pressure is shown in steps of 10 mmHg.

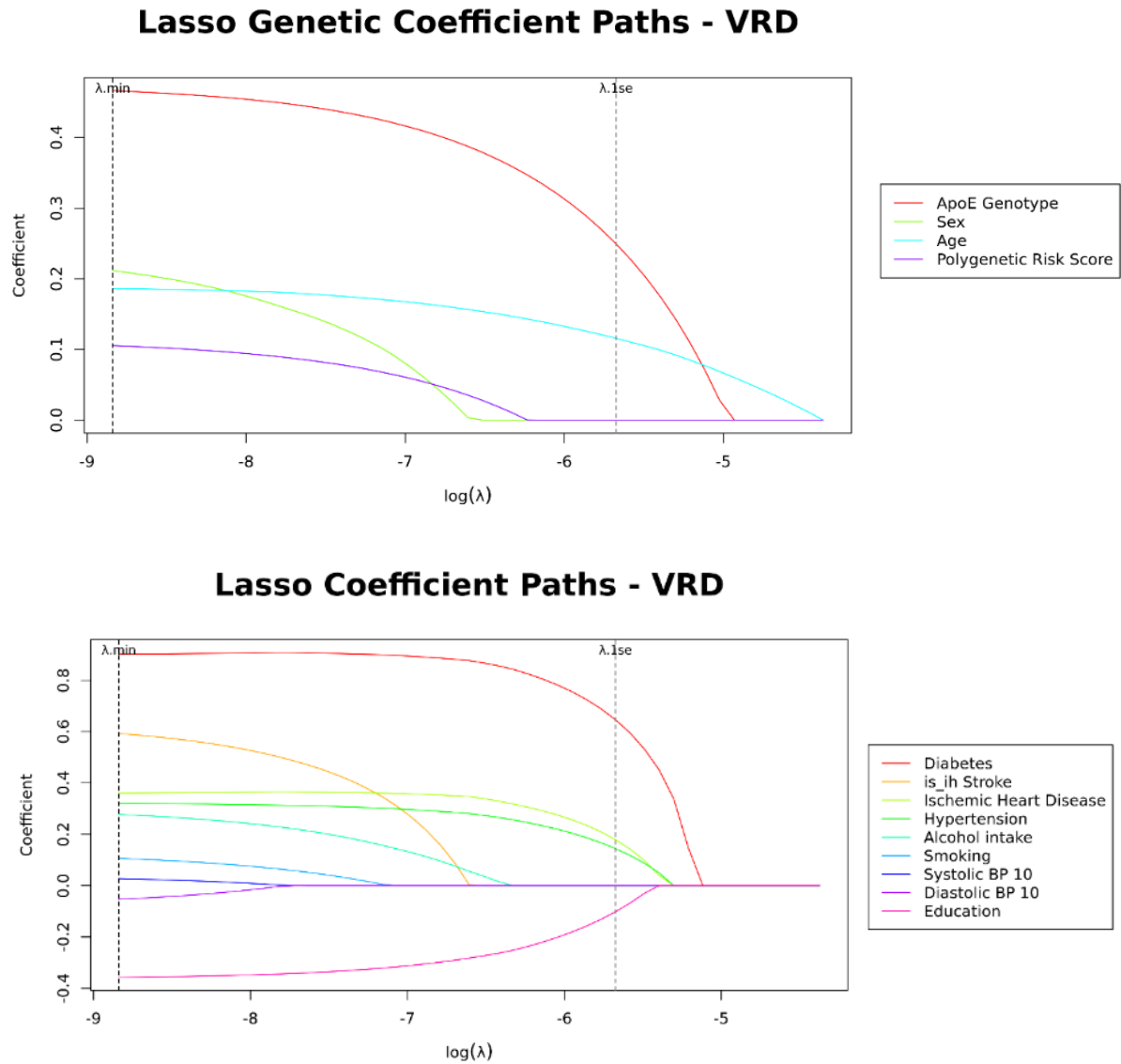


Figure A5. Upper and lower figures show Lasso regression for inherent and modifiable risk factors respectively for VRD. The figures show how the coefficients (β , y-axis) vary over different values for the penalty term (λ , x-axis). The maximum and minimum defensible penalty values are indicated with $\lambda_{\min}$ and λ_{1se} respectively using dotted lines. Covariates are colour coded. A covariate's coefficient which reaches zero at a lower penalty value is deemed less important and is more likely to be excluded.

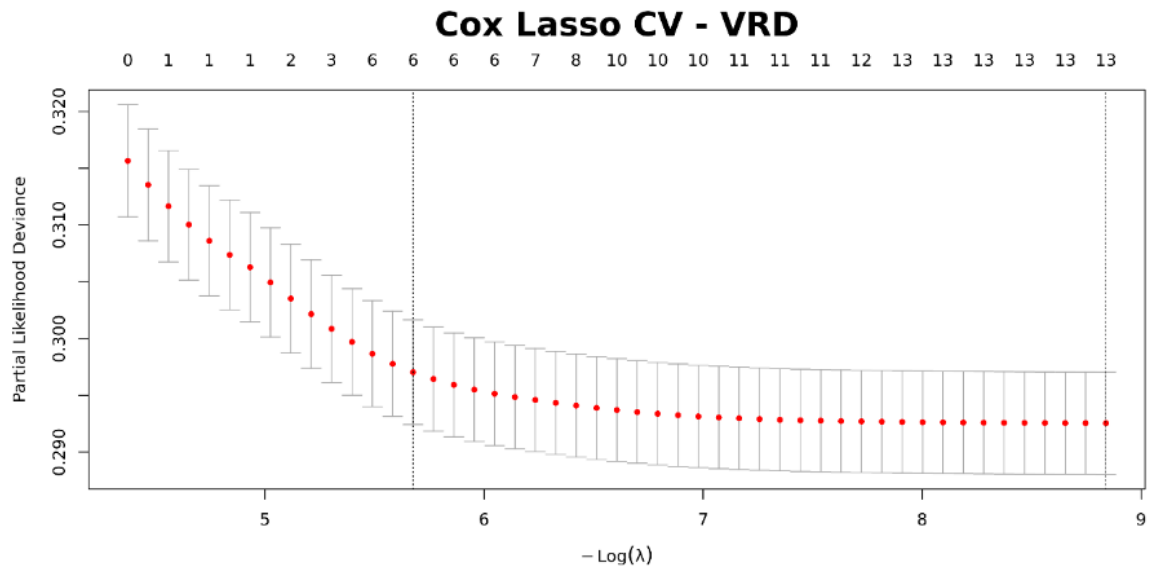


Figure A6. Lasso cross-validation of Cox models for VRD. The partial likelihood deviance (y-axis) as a function of penalty values (λ , x-axis) over a 10-fold cross-validation. The mean values of the cross-validation runs are shown in red, with corresponding 95% confidence intervals. At the top of the plot the total number of covariates included at the corresponding values of the penalty term are indicated.